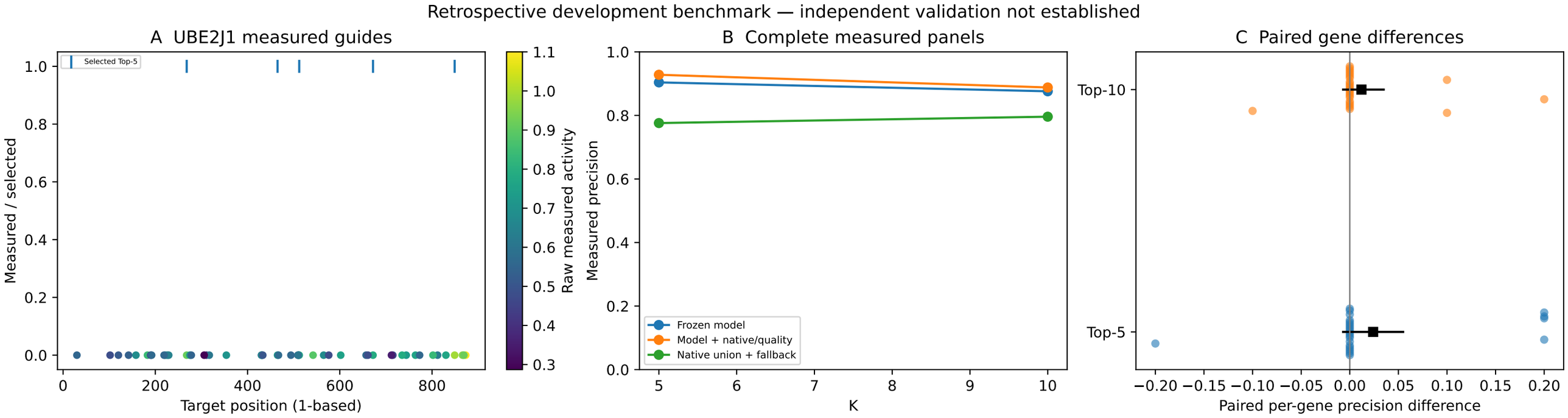
Retrospective development benchmark — independent validation not established
A UBE2J1 measured guides
Selected Top-5
Measured / selected
Target position (1-based)
Raw measured activity
B Complete measured panels
Measured precision
K
Frozen model
Model + native/quality
Native union + fallback
C Paired gene differences
Top-10
Top-5
Paired per-gene precision difference